STensor/DiffForm (Experimental)

Setup

In[1]:=
```
<< mGRG`STensor`
```

In[2]:=
```
$PreRead =
  ReplaceAll[#, expr_String :> StringReplace[expr, "SP" → "mGRG`STensor`Private"]] &;
```

In[3]:=
```
DefKind[Greek, Alphabet["Greek"]];
DefKind[Capital, ToUpperCase /@ Alphabet[]];
```

Define Differential Forms

DiffFormQ

In[ ]:=
```
DiffFormQ::usage
```

Out[ ]=
```
DiffFormQ[name] returns True if 'name'
  is a defined differential form, and False otherwise.
```

인자로 주어진 이름이 DefForm (또는 Fdefine)으로 정의된 것인가를 판단하는 질의함수이다.

In[ ]:=
```
Fdefine[f, 1];
DiffFormQ /@ {f, A}
```

Out[ ]=
```
{True, False}
```

DefForm, Fdefine

In[ ]:=
```
Fdefine::usage
```

Out[ ]=
```
Fdefine is an alias for DefForm.
```

In[ ]:=
```
DefForm::usage
```

Out[ ]=
```
DefForm[f, p, opts] defines a p-form f.
DefForm[f[indices...], p, sym, opts] defines a tensor-valued
  p-form with index symmetries given by the string 'sym'. The
  option PrintAs -> "str" can be used to set the output string.
```

Fdefine은 DefForm의 별칭이다.

새로운 IndexedObject인 미분 형식(Differential Form)을 정의한다. 인자로는 미분 형식의 이름, 차수(degree), 대칭 문자열이 필요하고, 옵션으로 출력을 위한 문자열이 있 다. 출력을 위한

문자열이 없으면 미분 형식의 이름이 출력을 위한 문자열로 사용된다. 또한, 대칭 문자열이 없으면 인덱스가 없는 보통의 미분 형식이 된다.

zero rank

인덱스 랭크가 0이고 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[α, 1];
SP`defineForm[α, 1, "", {}, {}];
{α, α[], α[la], α[la, lb]}
```
Out[∘]=
```
{α, α, α_a, α[la,lb]}
```

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[α]]
```
Out[∘]=
```
{0, GenSet[], {Latin}, {}, 1}
```

non-zero rank

랭크가 0이 아닌 미분 형식을 정의하려면 대칭 문자열을 입력해야 한다.

랭크 2이고 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[α[la, lb], 1];
{α, α[la], α[la, lb], α[lμ, la, lb], α[la, lb, lc, ld]}
```
Out[∘]=
```
{α, α[la], α_ab, α_μab, α[la,lb,lc,ld]}
```

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[α]]
```
Out[∘]=
```
{2, Nosymmetric, {Latin, Latin, Latin}, {-1}, 1}
```

랭크 3, 완전 대칭이고 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[A[la, lb, lc], 1, "3+", PrintAs → "α"];
{A, A[], A[la], A[la, lb], A[lb, la, lc], A[lμ, la, lb, lc], A[la, lb, lc, ld, le]}
```
Out[∘]=
```
{A, A[], A[la], A[la,lb], α_bac, α_μabc, A[la,lb,lc,ld,le]}
```

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{3, Symmetric, {Latin, Latin, Latin, Latin}, {-1}, 1}
```

랭크 3, 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[A[la, lb, uA], 1, PrintAs → "α"];
{A[la, uμ], A[la, lb, uA], A[lμ, la, lb, uA]}
```
Out[∘]=

$\{A[la, u\mu], \alpha_{ab}{}^{A}, \alpha_{\mu ab}{}^{A}\}$

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{3, Nosymmetric, {Latin, Latin, Latin, Capital}, {-1, -1, 1}, 1}
```

랭크 2, 완전 반대칭, 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[A[la, lb], 1, "2-", PrintAs → "α"];
{A[lb, la], A[lμ, lb, la]}
```
Out[∘]=

$\{\alpha_{ba}, \alpha_{\mu ba}\}$

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{2, Antisymmetric, {Latin, Latin, Latin}, {-1}, 1}
```

랭크 2이고 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[A[ua, ub], 1, PrintAs → "α"];
{A[ua, ub], A[lμ, ua, ub]}
```
Out[∘]=

$\{\alpha^{ab}, \alpha_{\mu}{}^{ab}\}$

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{2, Nosymmetric, {Latin, Latin, Latin}, {1}, 1}
```

(도입할 특별한 이유가 없어 보이는) 임의의 랭크인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[A[la], 1, "*", PrintAs → "α"];
{A, A[], A[lb], A[lμ, la], A[lμ, la, lb]}
```
Out[∘]=

$\{A, \alpha, \alpha_{b}, \alpha_{\mu a}, \alpha_{\mu ab}\}$

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{-1, Nosymmetric, {Latin, Latin}, {-1}, 1}
```

(도입할 특별한 이유가 없어 보이는) 임의의 랭크, 완전 반대칭, 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[A[la], 1, "*-", PrintAs → "α"];
{A, A[], A[la], A[lμ, la], A[lμ, la, lb]}
```
Out[∘]=
```
{A, α, α_a, α_μa, α_μab}
```

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{-1, Antisymmetric, {Latin, Latin}, {-1}, 1}
```

랭크 2, 반대칭, 차수가 1인 미분 형식을 정의한다:

In[∘]:=
```
Fdefine[A[la, lb], 1, "-ba"];
{A[lb, la], A[lμ, lb, la]}
```
Out[∘]=
```
{A_ba, A_μba}
```

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{2, GenSet[{Cycles[{{1, 2}}], -1}], {Latin, Latin, Latin}, {-1}, 1}
```

In[∘]:=
```
Fdefine[A[lμ, lν], 1, "-ba"];
{A[lν, lμ], A[lα, lν, lμ]}
```
Out[∘]=
```
{A_νμ, A_ανμ}
```

In[∘]:=
```
Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```
Out[∘]=
```
{2, GenSet[{Cycles[{{1, 2}}], -1}], {Latin, Greek, Greek}, {-1}, 1}
```

Check

In[∘]:=
```
Fdefine[A[la], 1, ""]
```
Msg: incompatible arguments for the length of indices

Out[∘]=
```
$Failed
```

잘못된 인덱스 개수의 확인은 출력될 때 자동적으로 진행된다. 그러나 오류 메세지는 프로그램 구조 때문에 출력되지 않는다. 오류 메세지를 출력하려면 SyntaxCheck 함수를 사용한다.

In[∘]:=
```
Fdefine[f, 0];
{f, f[], f[la]}
```
Out[∘]=
```
{f, f, f[la]}
```

In[ ]:=
```
f[la] // SyntaxCheck
```
Msg: invalid number of indices for **f** : {la}

Out[ ]=
```
f[la]
```

In[ ]:=
```
Fdefine[A, 1]
{A, A[la], A[la, lb], A[la, lb, lc]}
```
Out[ ]=

$\{A, A_a, A[la,lb], A[la,lb,lc]\}$

In[ ]:=
```
A[la, lb] // SyntaxCheck
```
Msg: invalid number of indices for **A** : {la, lb}

Out[ ]=
```
A[la,lb]
```

In[ ]:=
```
Fdefine[A[la], 2];
{A, A[la], A[la, lb], A[la, lb, lc]}
```
Out[ ]=

$\{A, A_a, A[la,lb], A_{abc}\}$

In[ ]:=
```
A[la, lb] // SyntaxCheck
```
Msg: invalid number of indices for A : {la, lb}

Out[ ]=
```
A[la,lb]
```

In[ ]:=
```
Fdefine[A[la, lb], 1]
{A, A[la], A[la, lb], A[la, lb, lc]}
```
Out[ ]=

$\{A, A[la], A_{ab}, A_{abc}\}$

In[ ]:=
```
A[la] // SyntaxCheck
```
Msg: invalid number of indices for A : {la}

Out[ ]=
```
A[la]
```

심볼 형태의 입력에 대해 (Indexed Object에 대한 오류를 확인하는) SyntaxCheck 함수는 반응하지 않는다:

In[ ]:=
```
A // SyntaxCheck
```
Out[ ]=
```
A
```

In[•]:=
```
A[]
```
Out[•]=
```
A[]
```

In[•]:=
```
A[] // SyntaxCheck
```
Msg: invalid number of indices for A

Out[•]=
```
A[]
```

UndefForm

In[•]:=
```
?A
```
Out[•]=

Symbol

Global`A

UpValue Definitions
```
mGRG`STensor`Private`getDegree[A] ^= 1

mGRG`STensor`Private`getDnupL[A] ^= {-1}

mGRG`STensor`Private`getGenSet[A] ^= Nosymmetric

mGRG`STensor`Private`getKindL[A] ^= {Latin, Latin, Latin}

mGRG`STensor`Private`getPrtStr[A] ^= A

mGRG`STensor`Private`getRank[A] ^= 2

mGRG`STensor`Private`getType[A] ^= DiffFormQ

IndexedObjectQ[A] ^= True

IndexedOperandQ[A] ^= True
```
Format Definitions
```
A /: MakeBoxes[A, StandardForm] := mGRG`STensor`Private`makeFormBox[A, 1, "A"]

A /: MakeBoxes[A[mGRG`STensor`Private`args$___], StandardForm] := mGRG`STensor`Pri
```
Full Name Global`A

정의된 미분 형식을 취소한다:

In[•]:=
```
UndefForm[A]
```

In[∘]:=
```
DiffFormQ[A]
```
Out[∘]=
```
False
```

In[∘]:=
```
?A
```
Out[∘]=
```
Symbol
Global`A
Full Name  Global`A
```

Differential Form Operators

XP: Exterior Product

XP type의 exterior product 연산자이다.

In[∘]:=
```
Fdefine[f, 0]; Fdefine[α, 1]; Fdefine[β, 2]; Fdefine[γ, 3]
```

0-form은 자동적으로 XP에서 제거되고, 또한 순서도 자동적으로 조정된다.

In[∘]:=
```
{XP[f, f], XP[f, α], XP[β, α], XP[α, α], XP[β, β]}
```
Out[∘]=
```
{f², f α, α∧β, 0, β∧β}
```

Wedge로 입력할 수 있다. (참고: Wedge를 입력하려면 "ESC, ^, ESC")

In[∘]:=
```
{f∧f, f∧α, β∧α, α∧α, β∧β}
```
Out[∘]=
```
{f², f α, α∧β, 0, β∧β}
```

DefaultKind의 공간 차원이 고려된다:

In[∘]:=
```
SetDimension[2]
```

In[∘]:=
```
{α∧f, α∧β}
```
Out[∘]=
```
{f α, 0}
```

In[∘]:=
```
ClearDimension[] (* default *)
```

Indexed DiffForm도 마찬가지이다.

In[∘]:=
```
Fdefine[f[la], 0]; Fdefine[α[la], 1]; Fdefine[β[la], 2]; Fdefine[γ[la], 3]
```

In[∘]:=
```
{f[la] ∧ α[lb], β[la] ∧ α[lb], α[lb] ∧ α[la], β[la] ∧ β[lb]}
```
Out[∘]=

$\{f_a\, \alpha_b,\ \alpha_b \wedge \beta_a,\ -\alpha_a \wedge \alpha_b,\ \beta_a \wedge \beta_b\}$

Indexd form에 대한 자동적인 단순화는 이루어지지 않는다:

In[∘]:=
```
α[ua] ∧ α[la]
```
Out[∘]=

$-\alpha_a \wedge \alpha^a$

In[∘]:=
```
SetDimension[2]
```

In[∘]:=
```
{α[la] ∧ f[lb], α[lb] ∧ β[la]}
```
Out[∘]=

$\{f_b\, \alpha_a,\ 0\}$

In[∘]:=
```
(α[la] + f[la]) ∧ β[ua]
```
Out[∘]=

$f_a\, \beta^a$

In[∘]:=
```
ClearDimension[] (* default *)
```

IP: Interior Product

LD type의 interior product 연산자이다.

In[∘]:=
```
Tdefine[v, 1];
Fdefine[f, 0]; Fdefine[α, 1]; Fdefine[β, 2]; Fdefine[γ, 3]
```

In[∘]:=
```
IP[v, #] & /@ {f, α, β, γ}
```
Out[∘]=

$\{\iota_v f,\ \iota_v \alpha,\ \iota_v \beta,\ \iota_v \gamma\}$

$$\iota_v\, \iota_v\, \omega = 0$$

In[∘]:=
```
{IP[v, IP[v, α]], IP[v, IP[v, β]], IP[v, IP[v, γ]]} (* α is 1-form *)
```
Out[∘]=

$\{\iota_v \iota_v \alpha,\ 0,\ 0\}$

0-form과 $p \geq 1$인 p-form의 곱인 경우 0-form은 자동적으로 IP에서 제거된다:

In[∘]:=
```
{IP[v, f], IP[v, f α]}
```
Out[∘]=

$\{\iota_v f,\ f\, \iota_v \alpha\}$

선형 연산자이다:

In[∘]:=
```
IP[v, a XD[α] + b β]
```
Out[∘]=

$b\,\iota_v\beta + a\,\iota_v d\alpha$

공식:

$$\iota_v\,\alpha\wedge\beta = (\iota_v\,\alpha)\wedge\beta + (-1)^p\,\alpha\wedge(\iota_v\,\beta)$$

In[∘]:=
```
IP[v, #] & /@ {f∧f, f∧α, β∧α, β∧β}
```
Out[∘]=

$\{\iota_v f^2,\ f\,\iota_v\alpha,\ \beta\,\iota_v\alpha - \alpha\wedge\iota_v\beta,\ 2\,\iota_v\beta\wedge\beta\}$

XD: Exterior Derivative

XD type의 exterior derivative 연산자이다.

In[∘]:=
```
Fdefine[f, 0]; Fdefine[α, 1]; Fdefine[β, 2]; Fdefine[γ, 3]
```

선형 미분 연산자이다.

In[∘]:=
```
{XD[f], XD[-α], XD[a β]}
```
Out[∘]=

$\{df,\ -d\alpha,\ a\,d\beta + da\wedge\beta\}$

Indexed DiffForm에 대해서도 마찬가지이다.

In[∘]:=
```
Fdefine[f[la], 0]; Fdefine[α[la], 1]; Fdefine[β[la], 2]; Fdefine[γ[la], 3]
```

In[∘]:=
```
{XD[f[la]], XD[-α[la]], XD[a β[la]]}
```
Out[∘]=

$\{df_a,\ -d\alpha_a,\ a\,d\beta_a + da\wedge\beta_a\}$

In[∘]:=
```
XD[β[la] ∧ α[ua]]
```
Out[∘]=

$d\alpha^a\wedge\beta_a - \alpha^a\wedge d\beta_a$

In[∘]:=
```
XD[α[la] ∧ α[ua]]
```
Out[∘]=

$-\alpha_a\wedge d\alpha^a + \alpha^a\wedge d\alpha_a$

Power, Log와 같은 ScalarFunction의 XD:

In[∘]:=
```
Fdefine[f, 0]
```

In[∘]:=
```
XD[f^2]
```
Out[∘]=

df^2

In[∘]:=
```
XD[Log[f]]
% // FullForm
```
Out[∘]=

dLog[f]

Out[∘]//FullForm=

```
XD[Log[f]]
```

In[∘]:=
```
XD[Tscalar[f]]
% // FullForm
```
Out[∘]=

df

Out[∘]//FullForm=

```
XD[f]
```

LD: Lie Derivative

텐서를 위한 LD 연산자는 미분 형식에 대해서도 동일하게 작용한다:

In[∘]:=
```
Tdefine[v, 1];
Fdefine[f[la], 0]; Fdefine[α[la], 1]; Fdefine[β[la], 2]
```

In[∘]:=
```
{LD[v, f[la]], LD[v, α[ua]], LD[v, β[1]]}
```
Out[∘]=

$\{\mathcal{L}_v f_a, \mathcal{L}_v \alpha^a, \mathcal{L}_v \beta^1\}$

In[∘]:=
```
{LD[v, α[la] ∧ β[lb]], LD[v, XD[α[la]]]}
```
Out[∘]=

$\{\mathcal{L}_v \alpha_a \wedge \beta_b + \alpha_a \wedge \mathcal{L}_v \beta_b, \mathcal{L}_v d\alpha_a\}$

선형 연산자이다.

In[∘]:=
```
SetAttributes[{c1, c2}, Constant]
```

In[∘]:=
```
LD[v, c1 XD[α[la]] + c2 β[la]]
```
Out[∘]=

$c1\, \mathcal{L}_v d\alpha_a + c2\, \mathcal{L}_v \beta_a$

LDtoXDRule

공식:

$$\mathcal{L}_v\,\omega = d\,(\iota_v\,\omega) + \iota_v\, d\omega$$

In[∘]:=
```
Tdefine[ξ[ua]];
Fdefine[ω[ua], 2]
```

In[∘]:=
```
LD[ξ, ω[ua]]
% /. LDtoXDRule[]
```

Out[∘]=

$\mathcal{L}_\xi \omega^a$

Out[∘]=

$\iota_\xi d\omega^a + d\,\iota_\xi \omega^a$

HodgeStar

In[∘]:=
```
HodgeStar::usage
```

Out[∘]=

HodgeStar[pForm] computes the Hodge dual of a p-form, resulting in an (n-p)-form, where n is the dimension of the manifold.

In[∘]:=
```
Tdefine[v, 1];
Fdefine[f, 0]; Fdefine[a, 1]; Fdefine[b, 2];
Fdefine[α[la], 1];
Fdefine[β[la], 1];
Fdefine[ω[ub], 2];
Fdefine[Ω[ua, ub], 2, "-ba"]
```

In[∘]:=
```
GetDimension[DefaultKind] = n; GetSig[DefaultKind] = s;
```

In[∘]:=
```
HodgeStar /@ {f, a, b}
```

Out[∘]=

$\{\star f, \star a, \star b\}$

In[∘]:=
```
HodgeStar /@ {α[la], β[la], ω[ub], Ω[la, lb]}
```

Out[∘]=

$\{\star\alpha_a, \star\beta_a, \star\omega^b, \star\Omega_{ab}\}$

공식: with GetSig = s

$$\star\star\,\omega = (-1)^{s+p\,(n+1)}\,\omega, \qquad \star^{-1} \equiv (-1)^{s+p\,(n+1)}\,\star$$

In[∘]:=
```
HodgeStar /@ HodgeStar /@ {α[la], β[la], ω[ub], Ω[la, lb]}
```

Out[∘]=

$\{(-1)^{1+n+s}\,\alpha_a, (-1)^{1+n+s}\,\beta_a, (-1)^{2\,(1+n)+s}\,\omega^b, (-1)^{2\,(1+n)+s}\,\Omega_{ab}\}$

```
In[]:= DegreeForm::usage
Out[]= DegreeForm[expr] returns the degree of a differential
         form expression. It correctly computes the degree for base forms,
         exterior products (XP), exterior derivatives (XD), interior
         products (IP), Hodge duals (HodgeStar), and codifferentials (CoXD).
```

```
In[]:= DegreeForm /@ (HodgeStar /@ {α[la], β[la], Ω[la, lb]})
Out[]= {-1 + n, -1 + n, -2 + n}
```

```
In[]:= DegreeForm /@ (HodgeStar /@ {ω[la, lb, uc], ω[ub]})
Out[]= {n, -2 + n}
```

```
In[]:= SetDimension[2]
```

```
In[]:= ZeroDegreeQ::usage
Out[]= ZeroDegreeQ[expr] returns True if 'expr' is a 0-form (a
         scalar) or not a differential form expression, and False otherwise.
```

```
In[]:= ZeroDegreeQ /@ (HodgeStar /@ {α[la], β[la], Ω[la, lb]})
Out[]= {False, False, True}
```

```
In[]:= ZeroDegreeQ /@ (HodgeStar /@ {ω[la, lb, uc], ω[ub]})
Out[]= {False, True}
```

```
In[]:= ClearDimension[]
```

```
In[]:= ZeroDegreeQ /@ (HodgeStar /@ {α[la], β[la], ω[ua], Ω[la, lb]})
Out[]= {False, False, False, False}
```

CoXD

```
In[]:= CoXD::usage
Out[]= CoXD[pForm] represents the codifferential operator acting on a p-form.
```

In[∘]:=
```
Tdefine[v, 1]; Fdefine[f, 0]; Fdefine[a, 1]; Fdefine[b, 2];
Fdefine[α[la], 1];
Fdefine[β[la], 1];
Fdefine[ω[ub], 2];
Fdefine[Ω[ua, ub], 2, "-ba"]
```

공식: s is the number of minuses in the signature of metric

$$\delta\omega = (-1)^{s+p\,n} * d * \omega = (-1)^{p} *^{-1} d\, *, \quad \delta\delta = 0$$

In[∘]:=
```
GetDimension[DefaultKind] = n; GetSig[DefaultKind] = s;
```

In[∘]:=
```
CoXD[ω[ua]]
% /. CoXDRule[]
```

Out[∘]=

$\delta\omega^{a}$

Out[∘]=

$(-1)^{2n+s} *d*\omega^{a}$

In[∘]:=
```
CoXD /@ {f, a, b}
```

Out[∘]=

$\{0, \delta a, \delta b\}$

In[∘]:=
```
CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]}
```

Out[∘]=

$\{\delta\alpha_{a}, \delta\beta_{a}, \delta\omega_{a}, \delta\Omega_{ab}\}$

In[∘]:=
```
CoXD /@ CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]}
```

Out[∘]=

$\{0, 0, 0, 0\}$

In[∘]:=
```
DegreeForm /@ (CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]})
```

Out[∘]=

$\{0, 0, 1, 1\}$

In[∘]:=
```
ZeroDegreeQ /@ (CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]})
```

Out[∘]=

{True, True, False, False}

Differential Form Operations

ApplyXD

In[5]:=
```
ApplyXD::usage
```

Out[5]=
```
ApplyXD[expr] explicitly applies the exterior
  derivative XD to an expression by expanding it in terms of
  basis derivatives BD with respect to the defined coordinates.
```

In[6]:=
$$\text{expr} = \left\{\left(1-2\,\frac{m}{r}\right)^{1/2} \text{XD}[t],\ \left(1-2\,\frac{m}{r}\right)^{-1/2} \text{XD}[r],\ r\,\text{XD}[\theta],\ r\,\text{Sin}[\theta]\,\text{XD}[\phi]\right\}$$

```
% // ApplyXD
```

Out[6]=
$$\left\{\sqrt{1-\frac{2\,m}{r}}\ dt,\ \frac{dr}{\sqrt{1-\frac{2\,m}{r}}},\ r\,d\theta,\ r\,\text{Sin}[\theta]\,d\phi\right\}$$

Out[7]=
$$\left\{d\sqrt{1-\frac{2\ m}{r}} \wedge dt,\ d\frac{1}{\sqrt{1-\frac{2\ m}{r}}} \wedge dr,\ dr\wedge d\theta,\ \text{Sin}[\theta]\,dr\wedge d\phi - r\,d\phi\wedge d\text{Sin}[\theta]\right\}$$

In[8]:=
```
SetCoordinates[{t, r, θ, ϕ}];
SetAttributes[{m}, Constant]
```

In[10]:=
```
expr // ApplyXD
```

Out[10]=
$$\left\{-\frac{m\,\hat{\partial}_r r\,dt\wedge dr}{\sqrt{1-\frac{2\,m}{r}}\ r^2} - \frac{m\,\hat{\partial}_\theta r\,dt\wedge d\theta}{\sqrt{1-\frac{2\,m}{r}}\ r^2} - \frac{m\,\hat{\partial}_\phi r\,dt\wedge d\phi}{\sqrt{1-\frac{2\,m}{r}}\ r^2},\ \frac{m\,\hat{\partial}_\theta r\,dr\wedge d\theta}{\left(1-\frac{2\,m}{r}\right)^{3/2} r^2} + \frac{m\,\hat{\partial}_\phi r\,dr\wedge d\phi}{\left(1-\frac{2\,m}{r}\right)^{3/2} r^2} - \frac{m\,\hat{\partial}_t r\,dt\wedge dr}{\left(1-\frac{2\,m}{r}\right)^{3/2} r^2},\right.$$
$$\hat{\partial}_r r\,dr\wedge d\theta + \hat{\partial}_t r\,dt\wedge d\theta - \hat{\partial}_\phi r\,d\theta\wedge d\phi,\ r\,\hat{\partial}_r\theta\,\text{Cos}[\theta]\,dr\wedge d\phi + \hat{\partial}_r r\,\text{Sin}[\theta]\,dr\wedge d\phi +$$
$$\left. r\,\hat{\partial}_t\theta\,\text{Cos}[\theta]\,dt\wedge d\phi + \hat{\partial}_t r\,\text{Sin}[\theta]\,dt\wedge d\phi + r\,\hat{\partial}_\theta\theta\,\text{Cos}[\theta]\,d\theta\wedge d\phi + \hat{\partial}_\theta r\,\text{Sin}[\theta]\,d\theta\wedge d\phi\right\}$$

In[11]:=
```
On[EvaluateBDFlag]
```

In[12]:=
```
expr // ApplyXD
```

Out[12]=
$$\left\{-\frac{m\,dt\wedge dr}{\sqrt{1-\frac{2\,m}{r}}\ r^2},\ 0,\ dr\wedge d\theta,\ \text{Sin}[\theta]\,dr\wedge d\phi + r\,\text{Cos}[\theta]\,d\theta\wedge d\phi\right\}$$

기본 설정으로 복귀한다:

In[13]:=
```
Off[EvaluateBDFlag]
```

In[14]:=
```
ClearCoordinates[]
```

CollectForm

In[15]:=
```
CollectForm::usage
```

Out[15]=

CollectForm[expr] collects terms in an expression that have the same differential form part, similar to how Collect works for standard algebraic expressions.

In[16]:=
```
Tdefine[ξ[ua]];
Fdefine[w1, 1]; Fdefine[w2, 2]; Fdefine[x[ua], 0]
```

In[18]:=
```
expr = IP[ξ, XD[w1]]
FtoC[expr, {la}] × XD[x[ua]]
```

Out[18]=

$\iota_{\xi}\mathbf{dw1}$

Out[19]=

$\nabla_a w1_b\, \mathbf{dx}^b\, \xi^a - \nabla_a w1_b\, \mathbf{dx}^a\, \xi^b$

In[20]:=
```
SumDum[%, {1, 3}]
CollectForm[%]
```

Out[20]=

$-\nabla_2 w1_1\, \mathbf{dx}^2\, \xi^1 + \nabla_1 w1_2\, \mathbf{dx}^2\, \xi^1 - \nabla_3 w1_1\, \mathbf{dx}^3\, \xi^1 + \nabla_1 w1_3\, \mathbf{dx}^3\, \xi^1 + \nabla_2 w1_1\, \mathbf{dx}^1\, \xi^2 - \nabla_1 w1_2\, \mathbf{dx}^1\, \xi^2 - \nabla_3 w1_2\, \mathbf{dx}^3\, \xi^2 + \nabla_2 w1_3\, \mathbf{dx}^3\, \xi^2 + \nabla_3 w1_1\, \mathbf{dx}^1\, \xi^3 - \nabla_1 w1_3\, \mathbf{dx}^1\, \xi^3 + \nabla_3 w1_2\, \mathbf{dx}^2\, \xi^3 - \nabla_2 w1_3\, \mathbf{dx}^2\, \xi^3$

Out[21]=

$\mathbf{dx}^3 \left(-\nabla_3 w1_1\, \xi^1 + \nabla_1 w1_3\, \xi^1 - \nabla_3 w1_2\, \xi^2 + \nabla_2 w1_3\, \xi^2\right) + \mathbf{dx}^2 \left(-\nabla_2 w1_1\, \xi^1 + \nabla_1 w1_2\, \xi^1 + \nabla_3 w1_2\, \xi^3 - \nabla_2 w1_3\, \xi^3\right) + \mathbf{dx}^1 \left(\nabla_2 w1_1\, \xi^2 - \nabla_1 w1_2\, \xi^2 + \nabla_3 w1_1\, \xi^3 - \nabla_1 w1_3\, \xi^3\right)$

In[22]:=
```
CoordRep[expr, 3]
```

Out[22]=

$\mathbf{dx}^3 \left(-\nabla_3 w1_1\, \xi^1 + \nabla_1 w1_3\, \xi^1 - \nabla_3 w1_2\, \xi^2 + \nabla_2 w1_3\, \xi^2\right) + \mathbf{dx}^2 \left(-\nabla_2 w1_1\, \xi^1 + \nabla_1 w1_2\, \xi^1 + \nabla_3 w1_2\, \xi^3 - \nabla_2 w1_3\, \xi^3\right) + \mathbf{dx}^1 \left(\nabla_2 w1_1\, \xi^2 - \nabla_1 w1_2\, \xi^2 + \nabla_3 w1_1\, \xi^3 - \nabla_1 w1_3\, \xi^3\right)$

In[23]:=
```
XP[w1, w2]
1/3! FtoC[%, {la, lb, lc}] × XD[x[ua]] ∧ XD[x[ub]] ∧ XD[x[uc]]
```

Out[23]=

$\mathbf{w1} \wedge \mathbf{w2}$

Out[24]=

$\frac{1}{2}\, w1_a\, w2_{bc}\, \mathbf{dx}^a \wedge \mathbf{dx}^b \wedge \mathbf{dx}^c$

In[25]:=
```
SumDum[%, {1, 3}] // TindexSort
% // CollectForm
```
Out[25]=

$w1_1\, w2_{23}\, dx^1 \wedge dx^2 \wedge dx^3 - w1_2\, w2_{13}\, dx^1 \wedge dx^2 \wedge dx^3 + w1_3\, w2_{12}\, dx^1 \wedge dx^2 \wedge dx^3$

Out[26]=

$(w1_1\, w2_{23} - w1_2\, w2_{13} + w1_3\, w2_{12})\, dx^1 \wedge dx^2 \wedge dx^3$

In[27]:=
```
CoordRep[XP[w1, w2], 3]
```
Out[27]=

$(w1_1\, w2_{23} - w1_2\, w2_{13} + w1_3\, w2_{12})\, dx^1 \wedge dx^2 \wedge dx^3$

In[28]:=
```
RemoveIndexedObject[x]
```

DegreeForm and ZeroDegreeQ

In[29]:=
```
DegreeForm::usage
```
Out[29]=

DegreeForm[expr] returns the degree of a differential form expression. It correctly computes the degree for base forms, exterior products (XP), exterior derivatives (XD), interior products (IP), Hodge duals (HodgeStar), and codifferentials (CoXD).

In[30]:=
```
Fdefine[ω[ua], 2]
```

In[31]:=
```
{ω[ua], ω[lb, ua], ω[lb, lc, ua]}
DegreeForm /@ %
ZeroDegreeQ /@ %%
```
Out[31]=

$\{\omega^a$, ω[lb,ua], $\omega_{bc}{}^a\}$

Out[32]=

{2, 0, 0}

Out[33]=

{False, True, True}

FtoC

In[34]:=
```
FtoC::usage
```
Out[34]=

FtoC[expr, {indices}] converts a differential form expression 'expr' into its corresponding antisymmetric indexed-tensor representation. The number of provided 'indices' must match the degree of the form.

In[35]:=
```
Tdefine[f[]]; Tdefine[ξ[ua]]; Tdefine[T[la, lb]];
Fdefine[w0, 0];
Fdefine[w1, 1];
Fdefine[w2, 2];
Fdefine[w3, 3];
Fdefine[w4, 4];
Fdefine[α[la], 1];
Fdefine[β[la], 1];
Fdefine[ω[ua], 2];
Fdefine[Ω[ua, ub], 2, "-ba"]
```

In[38]:=
```
SP`getGenSetOf /@ {ω[ub], ω[la, uc], ω[la, lb, uc]}
SP`getGenSetOf /@ {Ω[la, lb], Ω[la, lb, lc, ld]}
```

Out[38]=
```
{GenSet[], GenSet[], GenSet[{Cycles[{{1, 2}}], -1}]}
```

Out[39]=
```
{GenSet[{Cycles[{{1, 2}}], -1}],
 GenSet[{Cycles[{{1, 2}}], -1}, {Cycles[{{3, 4}}], -1}]}
```

In[40]:=
```
ω @@@ {{la}, {la, lb}, {la, lb, lc}, {la, lb, lc, ld}}
```

Out[40]=

$\{\omega_{a}, \omega[\mathrm{la,lb}], \omega_{abc}, \omega[\mathrm{la,lb,lc,ld}]\}$

p-form

In[41]:=
```
f T[la, lb]
FtoC[%, {}]
```

Out[41]=

$f\, T_{ab}$

Out[42]=

$f\, T_{ab}$

In[43]:=
```
ξ[ua] × T[lb, lc] w1
FtoC[%, {ld}]
```

Out[43]=

$\mathrm{w1}\, T_{bc}\, \xi^{a}$

Out[44]=

$T_{bc}\, \mathrm{w1}_{d}\, \xi^{a}$

In[45]:=
```
T[la, lb] w1
FtoC[%, {la}]
```

Out[45]=

$w1\ T_{ab}$

Msg: invalid indices: {la} for free indices {la, lb}

In[47]:=
```
T[la, lb] w1
FtoC[%, {ld}]
```

Out[47]=

$w1\ T_{ab}$

Out[48]=

$T_{ab}\ w1_{d}$

In[49]:=
```
T[la, lb] × ω[uc]
FtoC[%, {lc, ld}]
```

Out[49]=

$T_{ab}\ \omega^{c}$

Msg: invalid indices: {lc, ld} for free indices {la, lb, uc}

In[51]:=
```
T[la, lb] × ω[uc]
FtoC[%, {ld, le}]
```

Out[51]=

$T_{ab}\ \omega^{c}$

Out[52]=

$T_{ab}\ \omega_{de}{}^{c}$

In[53]:=
```
T[la, lb] w1 w2
FtoC[%, {lc}]
```

Out[53]=

$w1\ w2\ T_{ab}$

Msg: invalid number of indices {lc} for **w1 w2**

In[55]:=
```
T[la, lb] w1 w2
FtoC[%, {ub, lc, ld}]
```

Out[55]=

$w1\ w2\ T_{ab}$

Msg: invalid indices: {ub, lc, ld} for free indices {la, lb}

In[57]:=
```
T[la, lb] w1 w2
FtoC[%, {lc, ld, ue}]
```

Out[57]= $w1\, w2\, T_{ab}$

Out[58]= $T_{ab}\, w1_c\, w2_d{}^e$

In[59]:=
```
ω[ua]
FtoC[%, {lb, lc}]
```

Out[59]= ω^a

Out[60]= $\omega_{bc}{}^a$

In[61]:=
```
T[la, lb] w1 ω[uc] w2
FtoC[%, {ld, le, lf, lg, lh}]
```

Out[61]= $w1\, w2\, T_{ab}\, \omega^c$

Out[62]= $T_{ab}\, w1_d\, w2_{ef}\, \omega_{gh}{}^c$

In[63]:=
```
Ω[ua, ub]
FtoC[%, {lc, ld}]
```

Out[63]= Ω^{ab}

Out[64]= $\Omega_{cd}{}^{ab}$

XD

In[65]:=
```
Off[XDtoCDfrag]
```

In[66]:=
```
XD[x]
FtoC[%, {la}]
```

Out[66]= dx

Out[67]= $\partial_a x$

In[68]:=
```
On[XDtoCDfrag]
```

In[69]:=
```
XD[x]
FtoC[%, {la}]
```

Out[69]=

dx

Out[70]=

$\nabla_a x$

In[71]:=
```
TorsionFreeQ[CD] = False;
```

In[72]:=
```
XD[w1]
FtoC[%, {la, lb}]
```

Out[72]=

$dw1$

Out[73]=

$\partial_a w1_b - \partial_b w1_a$

In[74]:=
```
TorsionFreeQ[CD] = True;
```

In[75]:=
```
XD[w1]
FtoC[%, {la, lb}]
```

Out[75]=

$dw1$

Out[76]=

$\nabla_a w1_b - \nabla_b w1_a$

In[77]:=
```
T[la, lb] w1 XD[w1]
FtoC[%, {lc, ld, le}]
```

Out[77]=

$w1\, T_{ab}\, dw1$

Out[78]=

$\nabla_d w1_e\, T_{ab}\, w1_c - \nabla_e w1_d\, T_{ab}\, w1_c$

In[79]:=
```
T[la, lb] w1 w2 XD[ω[ua]]
FtoC[%, {la, lc, ld, le, lf, lg}]
```

Out[79]=

$w1\, w2\, T_{ab}\, d\omega^a$

Out[80]=

$\nabla_e \omega_{fg}{}^h\, T_{hb}\, w1_a\, w2_{cd} - \nabla_f \omega_{eg}{}^h\, T_{hb}\, w1_a\, w2_{cd} + \nabla_g \omega_{ef}{}^h\, T_{hb}\, w1_a\, w2_{cd}$

In[81]:=
```
T[la, lb] w1 ω[uc] × XD[Ω[ua, ub]]
FtoC[%, {la, lb, ld, le, lf, lg}]
```
Out[81]=
$$w1\, T_{ab}\, d\Omega^{ab}\, \omega^{c}$$
Out[82]=
$$\nabla_b\Omega_{de}{}^{hi}\, T_{hi}\, w1_a\, \omega_{fg}{}^{c} - \nabla_d\Omega_{be}{}^{hi}\, T_{hi}\, w1_a\, \omega_{fg}{}^{c} + \nabla_e\Omega_{bd}{}^{hi}\, T_{hi}\, w1_a\, \omega_{fg}{}^{c}$$

XP

In[83]:=
```
XP[XD[x], XD[y]]
FtoC[%, {la, lb}]
```
Out[83]=
$$dx\wedge dy$$
Out[84]=
$$-\nabla_a y\, \nabla_b x + \nabla_a x\, \nabla_b y$$

In[85]:=
```
XP[w1, w2]
FtoC[%, {la, lb, lc}]
```
Out[85]=
$$w1\wedge w2$$
Out[86]=
$$w1_c\, w2_{ab} - w1_b\, w2_{ac} + w1_a\, w2_{bc}$$

In[87]:=
```
T[la, lb] × XP[w1, w2]
FtoC[%, {lc, ld, le}]
```
Out[87]=
$$T_{ab}\, w1\wedge w2$$
Out[88]=
$$T_{ab}\, w1_e\, w2_{cd} - T_{ab}\, w1_d\, w2_{ce} + T_{ab}\, w1_c\, w2_{de}$$

LD

In[89]:=
```
LD[ξ, w0]
FtoC[%, {}]
```
Out[89]=
$$\mathcal{L}_{\xi} w0$$
Out[90]=
$$\nabla_a w0\, \xi^{a}$$

In[91]:=
```
LD[ξ, w1]
FtoC[%, {la}]
```
Out[91]=

$\mathcal{L}_{\xi}\mathbf{w1}$

Out[92]=

$\mathcal{L}_{\xi}\mathrm{w1}_{a}$

In[93]:=
```
LD[ξ, ω[ua]]
FtoC[%, {lb, lc}]
```
Out[93]=

$\mathcal{L}_{\xi}\boldsymbol{\omega}^{a}$

Out[94]=

$\mathcal{L}_{\xi}\omega_{bc}{}^{a}$

In[95]:=
```
LD[ξ, Ω[ua, ub]]
FtoC[%, {lc, ld}]
```
Out[95]=

$\mathcal{L}_{\xi}\boldsymbol{\Omega}^{ab}$

Out[96]=

$\mathcal{L}_{\xi}\Omega_{cd}{}^{ab}$

In[97]:=
```
LD[ξ, w1∧w2]
FtoC[%, {la, lb, lc}]
```
Out[97]=

$\mathbf{w1}\wedge\mathcal{L}_{\xi}\mathbf{w2} + \mathcal{L}_{\xi}\mathbf{w1}\wedge\mathbf{w2}$

Out[98]=

$\mathcal{L}_{\xi}\mathrm{w2}_{bc}\,\mathrm{w1}_{a} - \mathcal{L}_{\xi}\mathrm{w2}_{ac}\,\mathrm{w1}_{b} + \mathcal{L}_{\xi}\mathrm{w2}_{ab}\,\mathrm{w1}_{c} + \mathcal{L}_{\xi}\mathrm{w1}_{c}\,\mathrm{w2}_{ab} - \mathcal{L}_{\xi}\mathrm{w1}_{b}\,\mathrm{w2}_{ac} + \mathcal{L}_{\xi}\mathrm{w1}_{a}\,\mathrm{w2}_{bc}$

In[99]:=
```
LD[ξ, XD[x]]
FtoC[%, {la}]
```
Out[99]=

$\mathcal{L}_{\xi}\mathbf{dx}$

Out[100]=

$\mathcal{L}_{\xi}\nabla_{a}\mathrm{x}$

Check:

In[101]:=

```
LD[ξ, XD[x]]
% /. LDtoXDRule[]
FtoC[%, {la}]
```

Out[101]=

$\mathcal{L}_{\xi}\mathbf{dx}$

Out[102]=

$\mathbf{d}\,\iota_{\xi}\mathbf{dx}$

Out[103]=

$\nabla_a\xi^b\,\nabla_b x + \nabla_a\nabla_b x\,\xi^b$

In[104]:=

```
LD[ξ, CD[la, x]]
% // LDtoCD
```

Out[104]=

$\mathcal{L}_{\xi}\nabla_a x$

Out[105]=

$\nabla_a\xi^b\,\nabla_b x + \nabla_b\nabla_a x\,\xi^b$

IP

In[106]:=

```
IP[ξ, w1]
FtoC[%, {}]
```

Out[106]=

$\iota_{\xi}\mathbf{w1}$

Out[107]=

$w1_a\,\xi^a$

In[108]:=

```
IP[ξ, XP[w1, w2]]
FtoC[%, {la, lb}]
```

Out[108]=

$\mathbf{w2}\,\iota_{\xi}\mathbf{w1} - \mathbf{w1}\wedge\iota_{\xi}\mathbf{w2}$

Out[109]=

$w1_c\,w2_{ab}\,\xi^c - w1_b\,w2_{ac}\,\xi^c + w1_a\,w2_{bc}\,\xi^c$

In[110]:=
```
IP[ξ, XD[w1]]
FtoC[%, {la}]
```
Out[110]=
$\iota_\xi$dw1

Out[111]=
$-\nabla_a w1_b\, \xi^b + \nabla_b w1_a\, \xi^b$

In[112]:=
```
IP[ξ, XP[XD[x], XD[y]]]
FtoC[%, {la}]
```
Out[112]=
$-\iota_\xi$dy dx + $\iota_\xi$dx dy

Out[113]=
$\nabla_a y\, \nabla_b x\, \xi^b - \nabla_a x\, \nabla_b y\, \xi^b$

HodgeStar

참고: R. M. Wald (Appendix B.2, Problem 4.2.a)

$$\epsilon^{\mu_1 \ldots \mu_p\, \alpha_1 \ldots \alpha_{n-p}}\, \epsilon_{\mu_1 \ldots \mu_p\, \beta_1 \ldots \beta_{n-p}} = (-1)^s\, p!\, (n-p)!\, \delta^{[\alpha_1}_{\beta_1} \ldots \delta^{\alpha_{n-p}]}_{\beta_{n-p}}$$

$$(*\alpha)_{\beta_1 \ldots \beta_{n-p}} = \frac{1}{p!}\, \alpha^{\alpha_1 \ldots \alpha_p}\, \epsilon_{\alpha_1 \ldots \alpha_p \beta_1 \ldots \beta_{n-p}}$$

$$**\,\alpha = (-1)^{s+p(n-p)}\, \alpha = (-1)^{s+p(n+1)}\, \alpha$$

where s is the number of minuses appearing in the signature of g_{ab}.

In[114]:=
```
HodgeStar[XD[x]]
FtoC[%, {la}]
```
Out[114]=
*dx

Msg: Need to SetDimension[]

In[116]:=
```
SetDimension[3]; SetSig[0]
```

In[117]:=
```
HodgeStar[XD[x]]
FtoC[%, {la, lb}]
```
Out[117]=
*dx

Out[118]=
$\nabla_c x\, \epsilon^c{}_{ab}$

In[119]:=

```
HodgeStar[w1]
FtoC[%, {la, lb}]
```

Out[119]=

$\star w1$

Out[120]=

$\epsilon^{c}{}_{ab}\, w1_{c}$

In[121]:=

```
HodgeStar[ω[ua]]
FtoC[%, {lb}]
```

Out[121]=

$\star \omega^{a}$

Out[122]=

$\frac{1}{2}\, \epsilon^{cd}{}_{b}\, \omega_{cd}{}^{a}$

In[123]:=

```
HodgeStar[XP[XD[x], XD[y]]]
FtoC[%, {la}]
```

Out[123]=

$\star dx \wedge dy$

Out[124]=

$-\frac{1}{2}\, \nabla_{b}y\, \nabla_{c}x\, \epsilon^{bc}{}_{a} + \frac{1}{2}\, \nabla_{b}x\, \nabla_{c}y\, \epsilon^{bc}{}_{a}$

In[125]:=

```
HodgeStar[XP[w1, Ω[ub, uc]]]
FtoC[%, {}]
```

Out[125]=

$\star w1 \wedge \Omega^{bc}$

Out[126]=

$\frac{1}{6}\, \epsilon^{ade}\, w1_{e}\, \Omega_{ad}{}^{bc} - \frac{1}{6}\, \epsilon^{ade}\, w1_{d}\, \Omega_{ae}{}^{bc} + \frac{1}{6}\, \epsilon^{ade}\, w1_{a}\, \Omega_{de}{}^{bc}$

In[127]:=

```
HodgeStar[LD[ξ, XD[x]]]
FtoC[%, {la, lb}]
```

Out[127]=

$\star \mathcal{L}_{\xi} dx$

Out[128]=

$\epsilon^{c}{}_{ab}\, \mathcal{L}_{\xi} \nabla_{c}x$

In[129]:=
```
HodgeStar[LD[ξ, XD[ω[ua]]]]
FtoC[%, {}]
```
Out[129]=

$*\mathcal{L}_{\xi}\mathbf{d\omega}^{a}$

Out[130]=

$$\frac{1}{6}\in^{bcd}\mathcal{L}_{\xi}\nabla_{b}\omega_{cd}{}^{a}-\frac{1}{6}\in^{bcd}\mathcal{L}_{\xi}\nabla_{c}\omega_{bd}{}^{a}+\frac{1}{6}\in^{bcd}\mathcal{L}_{\xi}\nabla_{d}\omega_{bc}{}^{a}$$

In[131]:=
```
HodgeStar[IP[ξ, XD[x]]]
FtoC[%, {la, lb, lc}]
```
Out[131]=

$*\iota_{\xi}\mathbf{dx}$

Out[132]=

$\nabla_{d}x\in_{abc}\xi^{d}$

In[133]:=
```
HodgeStar[IP[ξ, XD[w1[]]]]
FtoC[%, {la, lb}]
```
Out[133]=

$*\iota_{\xi}\mathbf{dw1}$

Out[134]=

$\nabla_{c}\mathbf{w1}_{d}\in^{d}{}_{ab}\xi^{c}-\nabla_{c}\mathbf{w1}_{d}\in^{c}{}_{ab}\xi^{d}$

In[135]:=
```
HodgeStar[IP[ξ, XD[ω[ua]]]]
FtoC[%, {lb}]
```
Out[135]=

$*\iota_{\xi}\mathbf{d\omega}^{a}$

Out[136]=

$$\frac{1}{2}\nabla_{c}\omega_{de}{}^{a}\in^{de}{}_{b}\xi^{c}+\frac{1}{2}\nabla_{c}\omega_{de}{}^{a}\in^{cd}{}_{b}\xi^{e}-\frac{1}{2}\nabla_{c}\omega_{de}{}^{a}\in^{dc}{}_{b}\xi^{e}$$

In[137]:=
```
HodgeStar[IP[ξ, XD[Ω[ua, ub]]]]
FtoC[%, {lc}]
```
Out[137]=

$*\iota_{\xi}\mathbf{d\Omega}^{ab}$

Out[138]=

$$\frac{1}{2}\nabla_{d}\Omega_{ef}{}^{ab}\in^{ef}{}_{c}\xi^{d}+\frac{1}{2}\nabla_{d}\Omega_{ef}{}^{ab}\in^{de}{}_{c}\xi^{f}-\frac{1}{2}\nabla_{d}\Omega_{ef}{}^{ab}\in^{ed}{}_{c}\xi^{f}$$

In[139]:=

```
HodgeStar[x]
FtoC[%, {la, lb, lc}]
```

Out[139]=

$\star x$

Out[140]=

$x \in_{abc}$

In[141]:=

```
CoXD[w1] /. CoXDRule[]
```

Out[141]=

$-\star d\star w1$

In[142]:=

```
-HodgeStar[w1]
FtoC[%, {la, lb}]
```

Out[142]=

$-\star w1$

Out[143]=

$-\in^{c}{}_{ab}\, w1_{c}$

In[144]:=

```
-XD[HodgeStar[w1]]
FtoC[%, {la, lb, lc}]
```

Out[144]=

$-d\star w1$

Out[145]=

$-\nabla_c w1_d \in_{ab}{}^{d} + \nabla_b w1_d \in_{ac}{}^{d} - \nabla_a w1_d \in_{bc}{}^{d}$

In[146]:=

```
-HodgeStar[XD[HodgeStar[w1]]]
FtoC[%, {}]
```

Out[146]=

$-\star d\star w1$

Out[147]=

$$-\frac{1}{6}\nabla_a w1_b \in_{cd}{}^{b} \in^{acd} + \frac{1}{6}\nabla_a w1_b \in_{cd}{}^{b} \in^{cad} - \frac{1}{6}\nabla_a w1_b \in_{cd}{}^{b} \in^{cda}$$

In[148]:=
```
% /. EpsilonProductRule[]
% // Absorbg
```
Out[148]=
$-\nabla_a w1_b\, g^{ba}$

Out[149]=
$-\nabla^a w1_a$

In[150]:=
```
FtoC[CoXD[w1], {}]
```
Out[150]=
$-\nabla^a w1_a$

In[151]:=
```
ClearDimension[]; ClearSig[]
```

CoXD

In[152]:=
```
SetDimension[3]; SetSig[0]
```
In[153]:=
```
CoXD[XD[x]]
% /. CoXDRule[]
```
Out[153]=
δdx

Out[154]=
$-\star d\star dx$

In[155]:=
```
-HodgeStar[XD[HodgeStar[XD[x]]]]
FtoC[%, {}]
```
Out[155]=
$-\star d\star dx$

Out[156]=
$$-\frac{1}{6}\nabla_a\nabla_b x\,\epsilon_{cd}{}^{b}\,\epsilon^{acd}+\frac{1}{6}\nabla_a\nabla_b x\,\epsilon_{cd}{}^{b}\,\epsilon^{cad}-\frac{1}{6}\nabla_a\nabla_b x\,\epsilon_{cd}{}^{b}\,\epsilon^{cda}$$

In[157]:=
```
% /. EpsilonProductRule[]
% // Absorbg
```
Out[157]=
$-\nabla_a\nabla_b x\, g^{ba}$

Out[158]=
$-\nabla^a\nabla_a x$

In[159]:=
```
CoXD[XD[x]]
FtoC[%, {}]
```
Out[159]=
δdx

Out[160]=
$-\nabla^a\nabla_a x$

In[161]:=
```
CoXD[w1]
FtoC[%, {}]
```
Out[161]=
$\delta w1$

Out[162]=
$-\nabla^a w1_a$

In[163]:=
```
CoXD[w1]
% /. CoXDRule[]
```
Out[163]=
$\delta w1$

Out[164]=
$-\star d\star w1$

In[165]:=
```
-HodgeStar[XD[HodgeStar[w1]]]
FtoC[%, {}]
```
Out[165]=
$-\star d\star w1$

Out[166]=
$$-\frac{1}{6}\nabla_a w1_b \epsilon_{cd}{}^{b}\epsilon^{acd}+\frac{1}{6}\nabla_a w1_b \epsilon_{cd}{}^{b}\epsilon^{cad}-\frac{1}{6}\nabla_a w1_b \epsilon_{cd}{}^{b}\epsilon^{cda}$$

In[167]:=
```
% /. EpsilonProductRule[]
% // Absorbg
```
Out[167]=
$-\nabla_a w1_b\, g^{ba}$

Out[168]=
$-\nabla^a w1_a$

In[169]:=

```
CoXD[w2]
FtoC[%, {la}]
```

Out[169]=

$\delta w2$

Out[170]=

$-\nabla^{b} w2_{ba}$

In[171]:=

```
CoXD[w2]
% /. CoXDRule[]
```

Out[171]=

$\delta w2$

Out[172]=

$*d*w2$

In[173]:=

```
HodgeStar[XD[HodgeStar[w2]]]
FtoC[%, {la}]
```

Out[173]=

$*d*w2$

Out[174]=

$$\frac{1}{4} \nabla_{b} w2_{cd} \in_{e}{}^{cd} \in^{be}{}_{a} - \frac{1}{4} \nabla_{b} w2_{cd} \in_{e}{}^{cd} \in^{eb}{}_{a}$$

In[175]:=

```
% /. EpsilonProductRule[]
% // Absorbg
% // TindexSort
```

Out[175]=

$$\frac{1}{2} \nabla_{b} w2_{ac}\, g^{cb} - \frac{1}{2} \nabla_{b} w2_{ca}\, g^{cb}$$

Out[176]=

$$\frac{1}{2} \nabla^{b} w2_{ab} - \frac{1}{2} \nabla^{b} w2_{ba}$$

Out[177]=

$\nabla_{b} w2_{a}{}^{b}$

In[178]:=
```
CoXD[w3]
FtoC[%, {la, lb}]
```
Out[178]=
$\delta w3$

Out[179]=
$-\nabla^{c} w3_{cab}$

In[180]:=
```
CoXD[w3]
% /. CoXDRule[]
```
Out[180]=
$\delta w3$

Out[181]=
$-\star d \star w3$

In[182]:=
```
-HodgeStar[XD[HodgeStar[w3]]]
FtoC[%, {la, lb}]
```
Out[182]=
$-\star d \star w3$

Out[183]=
$-\frac{1}{6} \nabla_c w3_{def} \in^{c}{}_{ab} \in^{def}$

In[184]:=
```
% /. EpsilonProductRule[]
% // Absorbg
% // TindexSort
```
Out[184]=
$-\frac{1}{6} \nabla_c w3_{abd} g^{cd} + \frac{1}{6} \nabla_c w3_{adb} g^{cd} + \frac{1}{6} \nabla_c w3_{bad} g^{cd} - \frac{1}{6} \nabla_c w3_{bda} g^{cd} - \frac{1}{6} \nabla_c w3_{dab} g^{cd} + \frac{1}{6} \nabla_c w3_{dba} g^{cd}$

Out[185]=
$-\frac{1}{6} \nabla^{c} w3_{abc} + \frac{1}{6} \nabla^{c} w3_{acb} + \frac{1}{6} \nabla^{c} w3_{bac} - \frac{1}{6} \nabla^{c} w3_{bca} - \frac{1}{6} \nabla^{c} w3_{cab} + \frac{1}{6} \nabla^{c} w3_{cba}$

Out[186]=
$-\nabla_c w3_{ab}{}^{c}$

공식:

$$(\delta\omega)_{a...b} = -\nabla^{p} \omega_{pa...b}, \qquad \nabla_a \text{ : torsion - free CD}$$

Laplace-deRham operator

$$\Delta = -(\delta\, d + d\, \delta)$$

In[187]:=
```
Δ[expr_] := -(CoXD[XD[expr]] + XD[CoXD[expr]])
```

In[188]:=
```
Δ[w0]
FtoC[%, {}]
```
Out[188]=
$-\delta \mathrm{d}w0$

Out[189]=
$\nabla^{a}\nabla_{a}w0$

In[190]:=
```
Δ[w1]
FtoC[%, {la}]
```
Out[190]=
$-\delta \mathrm{d}w1 - \mathrm{d}\delta w1$

Out[191]=
$\nabla_{a}\nabla_{b}w1^{b} - \nabla^{b}\nabla_{a}w1_{b} + \nabla^{b}\nabla_{b}w1_{a}$

In[192]:=
```
CoXD[ω[ua]]
FtoC[%, {lb}]
```
Out[192]=
$\delta\omega^{a}$

Out[193]=
$-\nabla^{c}\omega_{cb}{}^{a}$

In[194]:=
```
XP[XD[x], XD[y]]
% // CoXD
FtoC[%, {la}]
```
Out[194]=
$\mathrm{d}x\wedge\mathrm{d}y$

Out[195]=
$\delta\mathrm{d}x\wedge\mathrm{d}y$

Out[196]=
$\nabla_{b}y\,\nabla^{b}\nabla_{a}x - \nabla_{b}x\,\nabla^{b}\nabla_{a}y - \nabla_{a}y\,\nabla^{b}\nabla_{b}x + \nabla_{a}x\,\nabla^{b}\nabla_{b}y$

In[197]:=

```
XP[w1, Ω[ub, uc]]
% // CoXD
FtoC[%, {la, ld}]
```

Out[197]=

$w1 \wedge \Omega^{bc}$

Out[198]=

$\delta w1 \wedge \Omega^{bc}$

Out[199]=

$-\nabla^{e}\Omega_{de}{}^{bc}\, w1_{a} + \nabla^{e}\Omega_{ae}{}^{bc}\, w1_{d} - \nabla^{e}\Omega_{ad}{}^{bc}\, w1_{e} - \nabla^{e}w1_{e}\, \Omega_{ad}{}^{bc} + \nabla^{e}w1_{d}\, \Omega_{ae}{}^{bc} - \nabla^{e}w1_{a}\, \Omega_{de}{}^{bc}$

In[200]:=

```
LD[ξ, XD[x]]
% // CoXD
FtoC[%, {}]
```

Out[200]=

$\mathcal{L}_{\xi}\mathbf{d}x$

Out[201]=

$\delta\mathcal{L}_{\xi}\mathbf{d}x$

Out[202]=

$-\nabla^{a}\mathcal{L}_{\xi}\nabla_{a}x$

In[203]:=

```
LD[ξ, XD[ω[ua]]]
% // CoXD
FtoC[%, {lb, lc}]
```

Out[203]=

$\mathcal{L}_{\xi}\mathbf{d}\omega^{a}$

Out[204]=

$\delta\mathcal{L}_{\xi}\mathbf{d}\omega^{a}$

Out[205]=

$-\nabla^{d}\mathcal{L}_{\xi}\nabla_{b}\omega_{cd}{}^{a} + \nabla^{d}\mathcal{L}_{\xi}\nabla_{c}\omega_{bd}{}^{a} - \nabla^{d}\mathcal{L}_{\xi}\nabla_{d}\omega_{bc}{}^{a}$

In[206]:=

```
IP[ξ, XD[x]]
% // CoXD
FtoC[%, {}]
```

Out[206]=

$\iota_{\xi}\mathbf{d}x$

Out[207]=

0

Out[208]=

0

In[209]:=

```
IP[ξ, XD[w1]]
% // CoXD
FtoC[%, {}]
```

Out[209]=

$\iota_{\xi}\mathbf{dw1}$

Out[210]=

$\delta\,\iota_{\xi}\mathbf{dw1}$

Out[211]=

$\nabla_a w1_b \,\nabla^a \xi^b - \nabla_a w1_b \,\nabla^b \xi^a + \nabla^a \nabla_a w1_b \,\xi^b - \nabla^a \nabla_b w1_a \,\xi^b$

In[212]:=

```
IP[ξ, XD[ω[ua]]]
% // CoXD
FtoC[%, {lb}]
```

Out[212]=

$\iota_{\xi}\mathbf{d}\omega^a$

Out[213]=

$\delta\,\iota_{\xi}\mathbf{d}\omega^a$

Out[214]=

$\nabla_b \omega_{cd}{}^a \,\nabla^c \xi^d - \nabla_c \omega_{bd}{}^a \,\nabla^c \xi^d + \nabla_c \omega_{bd}{}^a \,\nabla^d \xi^c + \nabla^c \nabla_b \omega_{cd}{}^a \,\xi^d - \nabla^c \nabla_c \omega_{bd}{}^a \,\xi^d + \nabla^c \nabla_d \omega_{bc}{}^a \,\xi^d$

In[215]:=

```
IP[ξ, XD[Ω[ua, ub]]]
% // CoXD
FtoC[%, {lc}]
```

Out[215]=

$\iota_{\xi}\mathbf{d}\Omega^{ab}$

Out[216]=

$\delta\,\iota_{\xi}\mathbf{d}\Omega^{ab}$

Out[217]=

$\nabla_c \Omega_{de}{}^{ab} \,\nabla^d \xi^e - \nabla_d \Omega_{ce}{}^{ab} \,\nabla^d \xi^e + \nabla_d \Omega_{ce}{}^{ab} \,\nabla^e \xi^d + \nabla^d \nabla_c \Omega_{de}{}^{ab} \,\xi^e - \nabla^d \nabla_d \Omega_{ce}{}^{ab} \,\xi^e + \nabla^d \nabla_e \Omega_{cd}{}^{ab} \,\xi^e$

Coordinate Representation: CoordRep

In[218]:=

```
CoordRep::usage
```

Out[218]=

```
CoordRep[form, coSys] gives the coordinate representation
  of the differential 'form' using the coordinate system 'coSys'.
CoordRep[form, n] gives the representation in a generic n-dimensional
  space, with the dimension defaulting to that of the Latin kind.
```

In[219]:=

```
Tdefine[v, 1]; Tdefine[ξ[ua]];
Fdefine[w1, 1]; Fdefine[w2, 2]; Fdefine[x[ua], 0]
```

In[221]:=

```
IP[ξ, XD[w1]]
FtoC[%, {la}]
```

Out[221]=

$\iota_\xi \mathbf{dw1}$

Out[222]=

$-\nabla_a w1_b\, \xi^b + \nabla_b w1_a\, \xi^b$

In[223]:=

```
SumDum[% XD[x[ua]], {1, 3}] // CollectForm
```

Out[223]=

$\mathbf{dx}^3 \left(-\nabla_3 w1_1\, \xi^1 + \nabla_1 w1_3\, \xi^1 - \nabla_3 w1_2\, \xi^2 + \nabla_2 w1_3\, \xi^2\right) + \mathbf{dx}^2 \left(-\nabla_2 w1_1\, \xi^1 + \nabla_1 w1_2\, \xi^1 + \nabla_3 w1_2\, \xi^3 - \nabla_2 w1_3\, \xi^3\right) + \mathbf{dx}^1 \left(\nabla_2 w1_1\, \xi^2 - \nabla_1 w1_2\, \xi^2 + \nabla_3 w1_1\, \xi^3 - \nabla_1 w1_3\, \xi^3\right)$

In[224]:=

```
CoordRep[IP[ξ, XD[w1]], 3]
```

Out[224]=

$\mathbf{dx}^3 \left(-\nabla_3 w1_1\, \xi^1 + \nabla_1 w1_3\, \xi^1 - \nabla_3 w1_2\, \xi^2 + \nabla_2 w1_3\, \xi^2\right) + \mathbf{dx}^2 \left(-\nabla_2 w1_1\, \xi^1 + \nabla_1 w1_2\, \xi^1 + \nabla_3 w1_2\, \xi^3 - \nabla_2 w1_3\, \xi^3\right) + \mathbf{dx}^1 \left(\nabla_2 w1_1\, \xi^2 - \nabla_1 w1_2\, \xi^2 + \nabla_3 w1_1\, \xi^3 - \nabla_1 w1_3\, \xi^3\right)$

In[225]:=

```
XP[w1, w2]
FtoC[%, {la, lb, lc}]
```

Out[225]=

$\mathbf{w1} \wedge \mathbf{w2}$

Out[226]=

$w1_c\, w2_{ab} - w1_b\, w2_{ac} + w1_a\, w2_{bc}$

In[227]:=

$\frac{1}{3!}$ % XD[x[ua]] ∧ XD[x[ub]] ∧ XD[x[uc]]

Out[227]=

$\frac{1}{2}\, w1_a\, w2_{bc}\, \mathbf{dx}^a \wedge \mathbf{dx}^b \wedge \mathbf{dx}^c$

In[228]:=

```
SumDum[%, {1, 3}] // TindexSort // CollectForm
```

Out[228]=

$\left(w1_1\, w2_{23} - w1_2\, w2_{13} + w1_3\, w2_{12}\right) \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$

In[229]:=

```
CoordRep[XP[w1, w2], 3]
```

Out[229]=

$(w1_1\, w2_{23} - w1_2\, w2_{13} + w1_3\, w2_{12})\ \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$

In[230]:=

```
SetDimension[3]
```

In[231]:=

```
Fdefine[ω[ua], 2]
```

In[232]:=

```
ω[ua]
FtoC[ω[ua], {lb, lc}]
```

Out[232]=

$\boldsymbol{\omega}^{a}$

Out[233]=

$\omega_{bc}{}^{a}$

In[234]:=

```
1/DegreeForm[ω[ua]] % XP[XD[x[ub]], XD[x[uc]]]
```

Out[234]=

$\frac{1}{2}\ \mathbf{dx}^b \wedge \mathbf{dx}^c\ \omega_{bc}{}^{a}$

In[235]:=

```
% // SumDum // TindexSort
```

Out[235]=

$\mathbf{dx}^2 \wedge \mathbf{dx}^3\ \omega_{23}{}^{a} + \mathbf{dx}^1 \wedge \mathbf{dx}^3\ \omega_{13}{}^{a} + \mathbf{dx}^1 \wedge \mathbf{dx}^2\ \omega_{12}{}^{a}$

In[236]:=

```
CoordRep[ω[ua]]
```

Out[236]=

$\mathbf{dx}^2 \wedge \mathbf{dx}^3\ \omega_{23}{}^{a} + \mathbf{dx}^1 \wedge \mathbf{dx}^3\ \omega_{13}{}^{a} + \mathbf{dx}^1 \wedge \mathbf{dx}^2\ \omega_{12}{}^{a}$